# Pulse Wave Velocity in Atrial Fibrillation: Clinical Significance and Current Evidence, A Scoping Review

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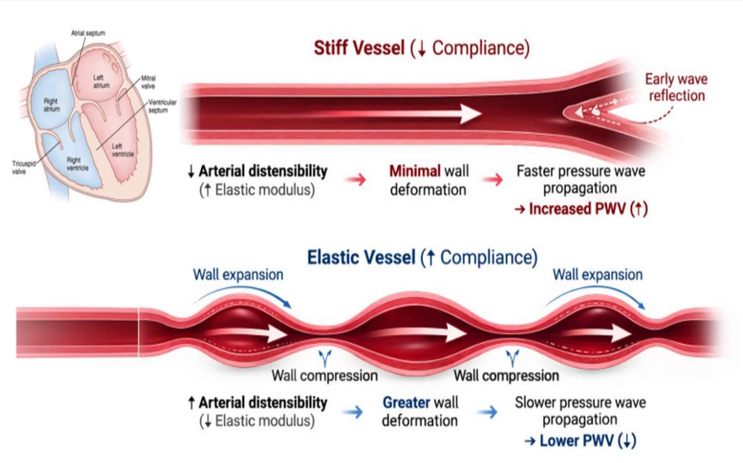
## BACKGROUND

- Atrial fibrillation (AF) is the most common sustained arrhythmia worldwide (>50 million people), and is associated with increased risk of stroke, heart failure, and mortality.
- Arterial stiffness reflects vascular aging and cumulative cardiovascular risk. Pulse wave velocity (PWV) is the gold-standard noninvasive measure of arterial stiffness.
- The clinical role of PWV across the spectrum of AF remains unclear.

### Physiological link between arterial stiffness and PWV

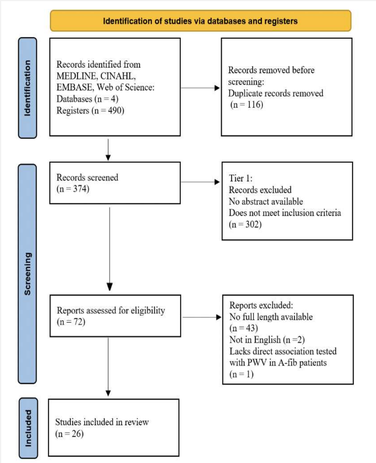



## OBJECTIVE

To conduct a scoping review evaluating the clinical significance of PWV in AF, including its role in:

- Prediction of incident AF
- Association between PWV and the presence of established AF
- AF burden, recurrence, and disease progression
- Post-operative AF risk stratification
- Clinical outcomes and prognosis in patients with AF
- Relationship between PWV and cardiovascular risk factors
- Clinical utility of PWV for risk stratification and patient management

## METHODS

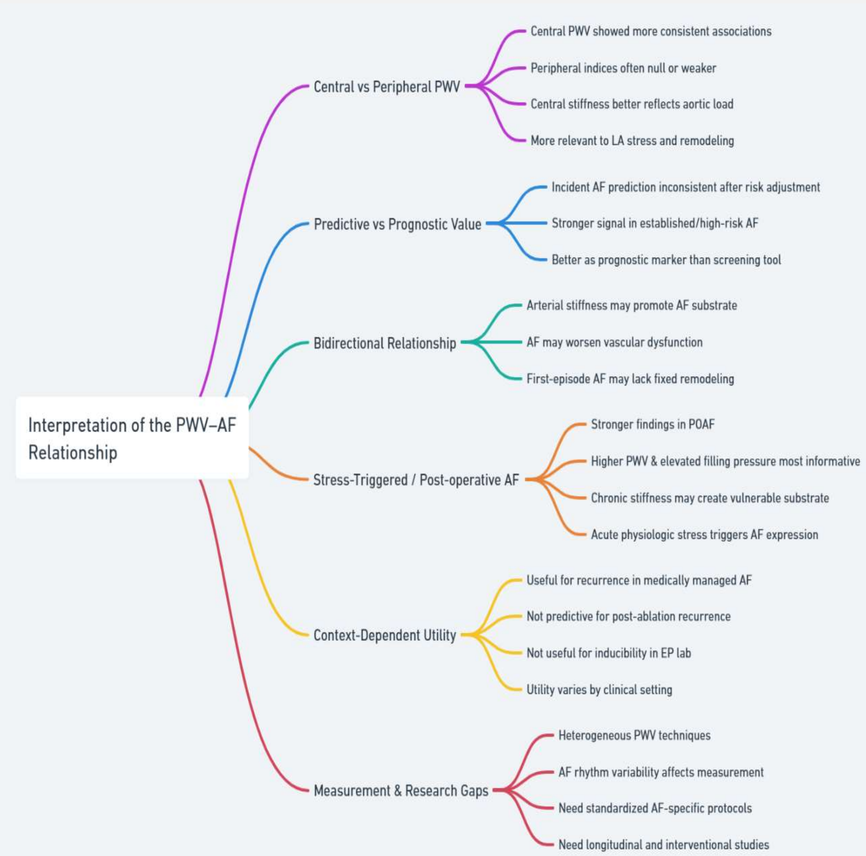



## RESULTS & DISCUSSION

### Summary of Evidence Linking PWV and AF

| AF Context | PWV Association | Clinical Meaning |
|---|---|---|
| Incident AF | Mixed | Limited screening value |
| Established AF | Consistent ↑ PWV | Marker of disease severity |
| Post-operative AF | Strong association | Perioperative risk marker |
| Prognosis | Higher PWV → worse outcomes | Risk stratification |
| Recurrence/Burden | Context dependent | Adjunct marker in select patients |

### Conceptual Interpretation of the PWV–AF Relationship




## CONCLUSIONS & KEY TAKEAWAYS

### Clinical Significance

- Elevated PWV is consistently associated with AF, particularly in patients with established disease or high cardiovascular risk.
- This relationship supports a mechanistic link between arterial stiffness, impaired ventricular–arterial coupling, and left atrial remodeling, which may promote susceptibility to AF.
- Central PWV demonstrates stronger associations with AF than peripheral PWV indices, likely reflecting its closer relationship to aortic loading conditions and cardiac hemodynamics.
- Elevated PWV may also reflect early atrial structural and functional abnormalities that precede clinically overt AF.

### Clinical Interpretation

- PWV appears more informative as a marker of prognosis than as a predictor of incident AF.
- Associations between PWV and AF appear strongest in established AF, POAF, and populations with high cardiovascular risk.
- In several studies, the relationship between PWV and incident AF was attenuated after adjustment for age, blood pressure, and comorbidities, suggesting that PWV reflects cumulative vascular risk.
- Evidence also suggests a bidirectional relationship, in which arterial stiffness may contribute to AF development while persistent AF may further impair vascular function.

### Clinical Application

PWV may provide additional risk stratification when interpreted alongside clinical risk scores and imaging markers of atrial cardiomyopathy.

- PWV may be particularly useful in stress-related clinical contexts such as POAF, where vascular stiffness may identify patients with increased susceptibility to arrhythmia.
- In medically managed AF, elevated PWV may help identify patients at higher risk of recurrence or disease progression.

## REFERENCES